

Water Quality Report - Station 85401000

2026-09-22

Station Information

Field	Value
Corpo d'agua	Arroio Francisco Alves
Ambiente	lotico
UF	RS
Entidade	FEPAM
Latitude	-29.702719
Longitude	-52.60335

Acronyms

- **CONAMA**: National Environment Council (Conselho Nacional do Meio Ambiente), the Brazilian body that sets the water-quality limits used in this screening.
- **ANA**: National Water and Sanitation Agency (Agencia Nacional de Aguas e Saneamento Basico), Brazil's federal water regulator and the source of the raw data.
- **RNQA**: National Water Quality Network (Rede Nacional de Qualidade da Agua), the monitoring program this data is published under.
- **FEPAM**: Rio Grande do Sul State Environmental Protection Foundation (Fundacao Estadual de Protecao Ambiental), the agency that operates this station.
- **NTU**: Nephelometric Turbidity Unit, the standard unit for turbidity.
- **NMP/100 mL**: Most Probable Number per 100 mL, the standard unit for bacterial counts such as E. coli.
- **Ambiente (lotico/lentico)**: lotico is flowing water (rivers, streams). Lentico is standing water (lakes, reservoirs). CONAMA 357 sets a stricter phosphorus limit for lentico environments, since still water eutrophies more easily.

Indicators

- **od** (Dissolved Oxygen): Oxygen available in the water for aquatic life. Low values indicate pollution or organic overload.
- **dbo** (Biochemical Oxygen Demand (BOD)): Oxygen consumed by microorganisms breaking down organic matter. High values indicate organic pollution such as sewage or effluents.
- **turbidez** (Turbidity): Water cloudiness caused by suspended particles. High values reduce light penetration and can indicate erosion or runoff.

- **fosforo_total** (Total Phosphorus): Nutrient commonly linked to agricultural or domestic runoff. Excess phosphorus causes eutrophication (excessive algae growth).
- **ecoli** (E. coli): Fecal-indicator bacteria. High counts signal sewage contamination and a health risk for contact or consumption.
- **iqa** (Water Quality Index (IQA)): A composite score from 0 to 100 combining several parameters into a single number; higher is better.

CONAMA 357/2005 class 2 screening

Indicator	Year	Value	Unit	Limit	Verdict
od	2018	-	-	-	no data
od	2019	8.38	mg/L	5.0	OK
od	2020	2.22	mg/L	5.0	Breach
dbo	2018	-	-	-	no data
dbo	2019	2.0	mg/L	5.0	OK
dbo	2020	5.0	mg/L	5.0	OK
turbidez	2018	-	-	-	no data
turbidez	2019	515.0	NTU	100.0	Breach
turbidez	2020	515.0	NTU	100.0	Breach
fosforo_total	2018	-	-	-	no data
fosforo_total	2019	-	-	-	no data
fosforo_total	2020	0.308	mg/L	0.1	Breach

Reference only (no CONAMA 357 limit)

Indicator	Year	Value	Unit
ecoli	2018	-	-
ecoli	2019	13004.8	NMP/100 mL
ecoli	2020	9476.4	NMP/100 mL
iqa	2018	-	-
iqa	2019	-	-
iqa	2020	-	-

Data Source

Raw measurements: ANA open data portal, Indicadores de Qualidade da Água (RNQA). Station operated by FEPAM.

Legal limits: - CONAMA 357/2005 art. 15 (classe 2): DBO 5 dias $\leq$ 5 mg/L
 - CONAMA 357/2005 art. 15 (classe 2): fosforo total ambiente lotico $\leq$ 0.1 mg/L
 - CONAMA 357/2005 art. 15 (classe 2): OD $\geq$ 5 mg/L
 - CONAMA 357/2005 art. 15 (classe 2): turbidez $\leq$ 100 UNT

Notes

- Values are yearly aggregates (mean/min/max), not individual samples. This is a screening, not a legal compliance verdict.
- The AMBIENTE mapping (1 = lotico, 2 = lentico) is inferred from water-body names, not from an official data dictionary.
- Data ends in 2021 for most indicators, 2020 for E. coli. Some years have gaps.
- E. coli and IQA are reported for reference only; no direct CONAMA 357 limit applies.
- This is a single-station extract of the same screening, not an independent lab analysis.

About This Report

Generated by `rs_water_quality`, an open-source Python and PostgreSQL pipeline that loads public ANA/RNQA water-quality data, screens it against CONAMA 357 limits, and produces this report automatically. Source code and methodology: <https://github.com/ar1vit0r/rs-water-quality>